Web Application Developer Intern

Company: NXP Semiconductors

Location: Bucharest, Romania

Contract: 1 year contract, part-time or full-time possibility

Seniority: Internship / Student role

Domain: Full-stack web development for internal engineering and PMO/PSO tools

Skill Summary for Dataset Extraction

Must-have skills identified from the job description:

- React
- JavaScript
- TypeScript
- Python
- Django
- HTTP
- REST
- SQL
- PostgreSQL
- Git
- Pull requests
- English communication
- Learning mindset

Nice-to-have skills identified from the job description:

- Django REST Framework
- Jest
- React Testing Library
- PyTest
- Docker
- CI/CD
- Pandas
- Data visualization
- Agile
- Technical documentation

Overview

NXP Semiconductors enables secure connections and infrastructure for a smarter world, advancing solutions that make lives easier, better and safer. As a world leader in secure connectivity solutions for embedded applications, NXP drives innovation in secure connected vehicles, end-to-end security and privacy, and smart connected solutions.

The company has 45,000 employees in more than 35 countries. More information can be found at www.nxp.com.

Your Team

Join the Tools team to help develop and improve an internal web application used by engineering and PMO/PSO stakeholders.

The role focuses on continuing the development of an internal tools platform with a React front end, a Django (Python) back end, and PostgreSQL databases.

The intern will design, build, and maintain features that improve engineering productivity and portfolio governance, working closely with product owners, stakeholders across AAES, and DevOps/IT partners using Agile methodologies.

Responsibilities

- Design, implement, test, and deploy web features across React, Django (Python), and PostgreSQL.
- Build RESTful APIs and integrate with internal services such as authentication, reporting, messaging, and data sources.
- Ensure backward compatibility for APIs and integrations.
- Design schemas, optimize queries, and apply indexing or partitioning strategies in PostgreSQL for reliability and scale.
- Write unit and integration tests using tools such as PyTest and Jest.
- Apply secure coding practices, perform code reviews, and maintain CI pipelines.
- Translate PSO/PMO workflows into intuitive user interfaces.
- Implement accessible and responsive UI components using React and modern state management.
- Partner with DevOps to containerize, deploy, monitor, and troubleshoot applications using tools such as Docker, GitHub pipelines, Azure DevOps pipelines, logging, and metrics.
- Maintain technical documentation and contribute to coding standards.
- Collaborate with PSO, VPMO, and RTE/STE communities to prioritize the roadmap and convert requirements into deliverables that support portfolio visibility and planning.
- Implement UI components and small features in React under mentorship.
- Fix UI bugs and improve accessibility and responsiveness.
- Add or adjust Django views, serializers, and REST endpoints.
- Write simple unit tests and assist with code reviews.
- Write and optimize basic PostgreSQL queries.
- Help with data validations and migrations.
- Contribute to automated testing with Jest for front end and PyTest for back end.
- Document changes and keep README notes up to date.
- Translate small workflow needs into incremental improvements that support portfolio dashboards and planning.

What You Will Learn

- Build production features in a mentored engineering environment.
- Learn API design basics, schema design, and performance fundamentals.
- Learn secure coding practices, code reviews, and CI/CD in an enterprise setup.
- Understand how PSO/PMO processes use tooling for KPIs, capacity, and portfolio visibility.

Must Have Skills

- Currently enrolled in a Bachelor's or Master's program in Computer Science, Software Engineering, or a related field.
- Coursework or projects using JavaScript or TypeScript and React.
- Good Python knowledge.
- Familiarity with web concepts such as HTTP and REST.

- Willingness to learn Django.
- Fundamental SQL knowledge, including joins, filtering, and simple aggregations.
- Comfortable using Git and collaborating through pull requests.
- Clear communication in English.
- Proactive and learning-oriented mindset.

Nice to Have Skills

- Exposure to Django REST Framework.
- Exposure to Jest, React Testing Library, or PyTest.
- Experience with basic Docker commands.
- Experience with any CI system.
- Familiarity with charts or visualization tools, such as Python pandas, for dashboards used by PSO/PMO.
- Ability to complete scoped tickets with mentor guidance.
- Ability to merge clean pull requests with adequate tests.
- Interest in improving test coverage and reducing small UI/API bugs.
- Interest in delivering incremental UX improvements and small API enhancements adopted by users.

Technology Stack

- Frontend: React, JavaScript, TypeScript, Jest, React Testing Library.
- Backend: Python, Django, Django REST Framework, RESTful APIs, PyTest.
- Database: PostgreSQL, SQL, schema design, indexing, migrations.
- DevOps and CI/CD: Docker, GitHub pipelines, Azure DevOps pipelines, logging, metrics.
- Collaboration: Git, pull requests, Agile methodologies, technical documentation.

Company Context

NXP Semiconductors N.V. makes products and environments safer, more sustainable, and more secure with innovative connectivity and edge processing solutions for a smarter world.

The company focuses on improving technologies and innovations that support society, business innovation, and the way people live and work.

What You Can Expect

- Contract: 1 year contract, with the possibility of part-time or full-time work.
- Compensation: salary package, lunch vouchers, and 25 vacation days.
- Work model: flexible work hours and work-from-home policy.
- Development opportunities: online and offline learning opportunities for core and professional skills.
- Office: located in a high-tech hub in Bucharest with access to public transport, restaurants, parks, relaxation areas, a library, cafeteria, and restaurant.
- Hiring process: online application, CV screening, initial phone or video conversation with Talent Acquisition, followed by business interviews.
- Company culture: social activities, free coffee, fresh fruit on Thursdays, and opportunities to participate in community support activities.

Security Note

The successful candidate may be responsible for security-related tasks. The assignment may be in scope of security certifications, so a conscious and reliable way of working is necessary.

Contact

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